



Paper ID : 261372

Printed Page: 1 of 1

Subject Code:MPC201T

Roll No:

MPHARM
(SEM. II) THEORY EXAMINATION 2025-26
ADVANCED SPECTRAL ANALYSIS

TIME: 3 HRS**M.MARKS: 75**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A**1. Attempt all questions in brief.****10 x 2 = 20**

a.	State the Woodward-Fieser Rule for 1,3-Butadienes.
b.	What is the significance of the Fingerprint Region in IR spectroscopy?
c.	Define Chemical Shift in ^{13}C NMR.
d.	What does the term NOESY stand for?
e.	Differentiate between Stokes and Anti-Stokes lines in Raman Spectroscopy.
f.	What is the principle of Flash Chromatography?
g.	Define the term Base Peak in Mass Spectrometry.
h.	What is the purpose of SDS-PAGE in electrophoresis?
i.	State Bragg's Law for X-ray diffraction.
j.	Mention the characteristic IR stretching frequency for a Carbonyl (C=O) group.

SECTION B**2. Attempt any two parts of the following:****2 x 10 = 20**

a.	Discuss the detailed Spectral Characterization (UV, IR, NMR, and Mass) of Aspirin. Provide the expected values for each technique.
b.	Explain the principle and instrumentation of Capillary Electrophoresis (CE). Discuss its advantages over conventional gel electrophoresis.
c.	What are Hyphenated Techniques? Describe the instrumentation and pharmaceutical applications of GC-MS (Gas Chromatography-Mass Spectrometry).

SECTION C**3. Attempt any five parts of the following:****7 x 5 = 35**

a.	Explain the fragmentation patterns of Alcohols and Amines in Mass Spectrometry.
b.	Describe the Southern Blotting technique and its significance in molecular analysis.
c.	Discuss the principle and applications of Supercritical Fluid Chromatography (SFC).
d.	Write a note on X-ray Powder Technique and its application in identifying drug polymorphs.
e.	Explain the difference between COSY and HETCOR in 2-D NMR with their structural importance.
f.	Estimate the λ_{max} of an α,β -unsaturated ketone using Woodward-Fieser rules with a suitable example.
g.	Discuss the interpretation of the NMR spectrum of Diazepam or Ephedrine